

# Concept Overview — Statistics & Probability

Covers all topics across the three assignments: **Unit 1 (Visualization)**, **Unit 2 (Central Tendency)**, **Unit 4 (Probability)**, and **Dispersion (Variance, Std Dev, IQR)**.

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## Unit 1: Data Visualization

### Line Diagram

Plots a variable (usually discrete, e.g., count of events) on the x-axis against frequency on the y-axis, connecting points with straight lines. Used to show the shape/trend of a discrete frequency distribution (e.g., number of accidents vs. number of drivers).

### Histogram

Used for **continuous grouped data** (class intervals). Bars are drawn touching each other (no gaps), where:

- Bar width = the width of the class interval
- Bar height = the frequency of that class

Unlike a bar chart, the bars touch because the underlying variable is continuous.

### Pie Diagram

Represents parts of a whole as sectors of a circle.

$$\text{Sector angle} = \frac{\text{value of category}}{\text{total of all values}} \times 360^\circ$$

Used for showing proportional/percentage composition (e.g., expenditure breakdown).

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## Unit 2: Measures of Central Tendency

### 1. Arithmetic Mean (simple & weighted)

Simple mean:

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n}$$

- $\bar{x}$  = the mean (average)
- $x_1, x_2, \dots, x_n$  = each individual data value
- $n$  = the number of values

**Weighted mean** — used when observations don't contribute equally (e.g., branch sales of different sizes):

$$\bar{x}_w = \frac{w_1x_1 + w_2x_2 + \dots + w_nx_n}{w_1 + w_2 + \dots + w_n}$$

- $x_i$  = each value being averaged (e.g., a growth percentage)
- $w_i$  = the weight attached to that value — how much it "counts," usually its size or importance (e.g., the sales amount tied to that growth rate)

Larger weights pull the average toward them, so a big branch's growth rate matters more than a small branch's.

## 2. Geometric Mean (GM)

Used for **rates of growth/decline, ratios, and percentages that compound over time** — e.g., growth in bad debts, a declining product price. A simple average is wrong here because growth rates multiply together over time rather than add.

$$GM = \sqrt[n]{x_1 \cdot x_2 \cdot x_3 \cdots x_n}$$

- $x_1, x_2, \dots, x_n$  = the individual values being averaged (often growth **factors**, like 1.10 for a 10% increase, rather than the raw percentage)
- $n$  = how many values there are
- The  $n$ -th root means: whatever number you get back, multiplying  $n$  copies of it together reproduces the product inside the root

**Key idea:** GM answers "what single constant rate, applied repeatedly, would produce the same overall change as the actual sequence of rates?" That's why it's the correct tool for average growth rate or average rate of decline.

To project forward  $t$  periods, raise the GM factor to that power:

$$\text{projected value} = GM^t$$

then convert the result back into a percentage change from the starting year.

## 3. Mode

The value that occurs most frequently. For **grouped/continuous data**, it's estimated with:

$$M = L + \left( \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \right) \times h$$

- $L$  = the lower boundary of the modal class (the class interval with the highest frequency)
- $f_1$  = the frequency (count) of the modal class itself
- $f_0$  = the frequency of the class interval right **before** the modal class
- $f_2$  = the frequency of the class interval right **after** the modal class
- $h$  = the width of each class interval

**Interpretation:** Mode identifies the most "typical"/common value and is less affected by extreme outliers than the mean — useful when data is skewed (e.g., ages where a few very old outliers pull the mean up).

## 4. Median

The middle value when data is arranged in order — the value that splits the distribution into a bottom half and top half.

**For grouped/continuous data:**

$$Min = L + \left( \frac{\frac{n}{2} - CF}{f} \right) \times h$$

- $L$  = the lower boundary of the median class (the class interval where the running total of frequency first reaches or passes half the total count)
- $n$  = the total number of observations (total frequency, all classes added together)
- $CF$  = the cumulative frequency **before** the median class (running total of all frequencies in classes below it)
- $f$  = the frequency (count) of the median class itself
- $h$  = the width of each class interval

**Interpretation:** Median is not affected much by outliers or skewness, which is why it's preferred over the mean for skewed distributions like income or age data with long tails.

**Finding missing frequencies given the median:** Plug the known median value into the formula above with the unknown frequency left as a variable, then solve algebraically — often combined with the fact that all frequencies must add up to the known total.

## Unit 4: Probability Fundamentals

### Classical Probability

$$P(A) = \frac{\text{nuer of favorable outcoes}}{\text{total nuer of euallly liely outcoes}}$$

## Combinatorics (counting outcomes)

Used heavily in card-drawing problems. The number of ways to choose items out of  $n$  items (order doesn't matter):

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

- $n$  = total number of items available
- $r$  = number of items being chosen
- (factorial) means multiply the number by every whole number below it down to 1 (e.g.,  $4! = 4 \times 3 \times 2 \times 1$ )

E.g., the probability of drawing a specific combination of cards = (number of ways to form the favorable combination)  $\div$  (number of ways to choose any 4 cards out of 52).

## Addition Rule (for "or")

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- $P(A \cup B)$  = probability that A **or** B (or both) happens
- $P(A \cap B)$  = probability that A **and** B both happen (the overlap)

Used for problems like "multiple of 2 or 5" — you subtract the overlap (numbers that are multiples of both) so you don't count those numbers twice.

## Joint, Marginal, and Conditional Probability

Given a joint probability table (like the manufacturer/defects or economists tables):

- **Joint probability**  $P(A \cap B)$  — probability of two things happening together, read directly from a single cell of the table.
- **Marginal probability**  $P(A)$  — probability of just one event by itself, found by adding up a whole row or column of the table.
- **Conditional probability** — the probability of one event *given that another has already happened*:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}$$

- $P(A | B)$  = probability of A, given that B is known to have happened
- This restricts your attention to only the cases where B occurred, then asks how often A also occurred within those cases

## Statistical Independence

Two events A and B are **independent** if knowing one tells you nothing about the other. This is true exactly when:

$$P(A \cap B) = P(A) \times P(B) \text{ equivalently } P(A | B) = P(A)$$

To check independence with real data, compare  $P(A | B)$  to plain  $P(A)$  — if they come out equal (or very close), the two events don't really influence each other (e.g., TV ownership vs. telephone subscription).

## Probability With/Without Replacement

- **Without replacement:** probabilities change after each draw because the pool you're drawing from shrinks and its makeup changes (the draws are not independent) — handle this with combinations or step-by-step conditional probabilities.
- **With replacement:** each draw is identical and doesn't affect the next one (the draws are independent) — simply multiply the probability of each individual draw together.

## Total Probability (mixture of sources)

When an outcome can happen through several different, non-overlapping paths (e.g., drawing from Bag 1 OR Bag 2, each equally likely to be picked):

$$P(A) = P(A | B_1)P(B_1) + P(A | B_2)P(B_2) + \dots$$

- $B_1, B_2, \dots$  = the different possible "paths" or sources (e.g., which bag was chosen)
  - $P(B_i)$  = probability of that path being the one used
  - $P(A | B_i)$  = probability of the outcome happening, given that path was used
  - One term is added for every possible path
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## Measures of Dispersion: Variance, Standard Deviation, IQR

Central tendency tells you where the data is centered; **dispersion** tells you how spread out it is. Two datasets can share the same mean but differ wildly in spread.

### Variance

The average of the squared distances between each data point and the mean. Squaring is used so that positive and negative deviations don't cancel out, and so bigger deviations count more heavily.

**Sample variance:**

$$s^2 = \frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \dots + (x_n - \bar{x})^2}{n - 1}$$

- $x_i$  = each individual data value
- $\bar{x}$  = the mean of the sample
- $(x_i - \bar{x})$  = that value's distance from the mean
- $n$  = number of data points in the sample
- $n - 1$  (instead of just  $n$ ) is used for a sample so the result is an unbiased estimate of the true population variance — this adjustment is called Bessel's correction

**For grouped/frequency data:**

$$s^2 = \frac{f_1(x_1 - \bar{x})^2 + f_2(x_2 - \bar{x})^2 + \dots + f_n(x_n - \bar{x})^2}{f_1 + f_2 + \dots + f_n}$$

- $x_i$  = the midpoint (or value) of each class
- $f_i$  = the frequency (count) of each class
- The denominator is simply the total of all frequencies

Variance is measured in **squared units** of the original data (e.g., "dollars squared"), which is hard to interpret directly — that's why standard deviation is usually reported instead.

## Standard Deviation

Simply the square root of the variance, which brings the measure back into the **same units** as the original data.

$$s = \sqrt{s^2} \text{ or } = \sqrt{s^2}$$

**Interpretation:** Standard deviation tells you the "typical" distance of a data point from the mean. A small standard deviation means the data clusters tightly around the mean; a large one means the data is spread widely. It's the most commonly used spread measure because it's in easy-to-interpret units and underlies many further concepts (z-scores, the normal distribution, confidence intervals).

**Shortcut formula** (avoids computing every individual deviation by hand):

$$s^2 = \frac{x_{21} + x_{22} + \dots + x_{2n}}{n} - \bar{x}^2$$

- The first term is the average of all the squared values
- $\bar{x}^2$  is the mean, squared

## Interquartile Range (IQR)

Measures the spread of just the **middle 50%** of the data, which makes it resistant to outliers (unlike variance/standard deviation, which are sensitive to extreme values because of the squaring step).

1.  $Q_1$  (**first quartile**): the value below which the smallest 25% of the data falls.
2.  $Q_3$  (**third quartile**): the value below which the smallest 75% of the data falls (equivalently, the top 25% lies above it).
3. 
$$Q = Q_3 - Q_1$$

For **grouped/continuous data**, quartiles are found the same way as the median, just targeting a quarter or three-quarters of the total instead of half:

$$Q = L + \left(\frac{\frac{n}{4} - CF}{f}\right) \times h \quad (= 1 \text{ or } 3)$$

- = which quartile you want (1 for  $Q_1$ , 3 for  $Q_3$ )
- $L, CF, f, h$  have the same meaning as in the median formula, but are located using the running total that crosses the quartile's target count ( $\frac{n}{4}$ ) instead of the median's halfway point ( $\frac{n}{2}$ )

**Interpretation & use:** IQR is often paired with the median (rather than mean/standard deviation) for skewed distributions, and is the basis of the **box-and-whisker plot** and the standard rule for spotting outliers:

$$\text{outlier if } x < Q_1 - 1.5 \times IQR \text{ or } x > Q_3 + 1.5 \times IQR$$

## Quick Reference Table

| Concept                 | Formula                                           | Used When                                         |
|-------------------------|---------------------------------------------------|---------------------------------------------------|
| Weighted Mean           | $\bar{x}_w = \frac{\sum w_i x_i}{\sum w_i}$       | Averaging quantities of unequal "size"/importance |
| Geometric Mean          | $GM = \sqrt[n]{x_1 x_2 \cdots x_n}$               | Averaging growth rates, ratios, % changes         |
| Mode (grouped)          | $L + \frac{f_1 - f_0}{2f_1 - f_0 - f_2} \times h$ | Most frequent/typical value, skewed data          |
| Median (grouped)        | $L + \frac{\frac{n}{2} - CF}{f} \times h$         | Middle value, robust to outliers                  |
| Conditional Probability | $P(A   B) = \frac{P(A \cap B)}{P(B)}$             | Probability given some condition holds            |

| Concept            | Formula                                      | Used When                                     |
|--------------------|----------------------------------------------|-----------------------------------------------|
| Independence check | $P(A \cap B) = P(A)P(B)$                     | Testing if two events affect each other       |
| Total Probability  | $P(A) = \sum_i P(A   B_i)P(B_i)$             | Outcome from multiple possible sources        |
| Variance (sample)  | $s^2 = \frac{\sum (x_i - \bar{x})^2}{n - 1}$ | Measuring average squared spread from mean    |
| Standard Deviation | $s = \sqrt{s^2}$                             | Typical distance from mean, in original units |
| IQR                | $Q_3 - Q_1$                                  | Spread of middle 50%, robust to outliers      |